Advisor: Bernardo Guimarães

Conditional Cash Transfers as Stabilization Tool: a HANK Approach

1. Model

The setup is a standard New Keynesian model with incomplete markets and labor supply risk. Time is discrete and runs from $t = 0$ to infinity. There are four types of agents in the economy: households, firms, labor unions, and the government.

Households are heterogeneous (Aiyagari 1994), workers and firms join in a formal-informal setup, and the rest of the economy follows the textbook New Keynesian model (Galí 2015). There is no aggregate uncertainty and the economy starts in the deterministic steady-state.

Households

Assume there exists a continuum $i \in [0, 1]$ of ex-ante identical households, but experiencing idiosyncratic shocks in labor productivity and discount factor.

Households enjoy a CRRA-GHH utility over consumption $c_{i,t}$ and worked hours $h_{i,t}$ (Greenwood, Hercowitz, and Huffman 1988)[1], which we define $u(\tilde{c}_{i,t}) = u(c_{i,t} - v(h_{i,t}))$ as

$$\max_{c_{i,t}, h_{i,t}} \; \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{\tilde{c}_{i,t}^{1-\frac{1}{\gamma}}}{1 - \frac{1}{\gamma}} \right], \quad \text{with } v(h) = \psi \frac{h^{1+\frac{1}{\varphi}}}{1 + \frac{1}{\varphi}}$$

where γ is the elasticity of intertemporal substitution (EIS) and φ is the Frisch elasticity of labor supply.

A household faces idiosyncratic labor productivity shocks $e_{i,t}$ with $\mathbb{E}[e_{i,t}] = 1$, and discount factor heterogeneity $\beta_i \in \{\beta_L, \beta_H\}$, where there is ω_I impatient households. They can work in the formal or informal sector, or be unemployed $s \in \{F, I, U\}$. Each sector has a specific productivity $\theta_{i,t}^s$, re-drawn on each new offer.

Formal workers earn labor income $(1 - \tau_\ell)\theta_{i,t}^F w_t e_{i,t} h_{i,t}$, where w_t is the formal wage rate and τ_ℓ is a fixed labor income tax rate. While, informal workers receive benefits $\theta_{i,t}^I w_t^I e_{i,t} h_{i,t}$,

[1] The GHH specification assumes zero income effect, so that the hours decision is separable from the consumption-savings problem.

where $w_t^I = \xi w_t$ and $\xi \in (0,1)$ is the wage gap. Let $y_t(s,e)$ be this non-financial income,

$$y_t(s,e,\theta) = \begin{cases} (1-\tau_t)\theta^F w_t h_t e, & \text{if formal;} \\ \theta^I w_t^I h_t e + T_t \cdot \mathbb{1}[\text{eligible}], & \text{if informal;} \\ T_t, & \text{if unemployed.} \end{cases}$$

We assume that formal wages are observable by the government, so that only informal workers and unemployed may be eligible to receive Bolsa Família (BF). We assume that informal households are eligible if $y_{i,t}(I,e,\theta) < \bar{y}$ and that the total benefit size is BF_t.

Households save into liquid assets $a_{i,t}$ with ex-ante real return $1+r_t = (1+i_{t-1})/(1+\pi_t)$, where i_t represents the nominal interest rate and π_t denotes inflation. They also receive dividends $d_{i,t}$ from firms, which we assume to be proportional to productivity $d_{i,t} = d_t \cdot e_{i,t}$[2]. Their budget constraint at time t reads:

$$c_{i,t} + a_{i,t} = (1+r_t)a_{i,t-1} + d_{i,t} + y_{i,t}(s,e).$$

Incomplete markets impose a borrowing constraint $a_{i,t} \geq \underline{a}$. Frictions in the labor market imply formal households delegate their labor supply decisions to labor unions, and thus take worked hours as given. While, informal ones choose hours to maximize their utility, thus, for each i,

$$\psi\,(h)^{1/\varphi} = \theta w^I e \quad \Leftrightarrow \quad h^I(e) = \left[\frac{\theta w^I e}{\psi}\right]^{\varphi}.$$

This generates a productivity cutoff for informal workers of

$$\theta w^I e \left(\frac{\theta w^I e}{\psi}\right)^{\varphi} < \bar{y} \quad \Leftrightarrow \quad e < e^* = \frac{(\psi^{\varphi}\bar{y})^{\frac{1}{1+\varphi}}}{\theta w^I}.$$

Such as Esteban-Pretel and Kitao (2021) and Finamor (2025), given a sector s, we assume that each worker faces an exogenous layoff probability δ_s, and receives a formal offer with probability π_F and an informal one with probability π_I. The worker accepts the offer with higher value.

[2]Dividends are proportional to productivity in order to increase inequality. We aim to perform robustness tests with other specifications.

The Bellman Equation for each individual i in sector $s \in \{F, I\}$ is

$$\begin{aligned}
V_t^s(a,e,\beta,\theta) = \max_{\tilde{c},a'} \Bigg\{ & \frac{\tilde{c}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \Big[\delta_s V_{t+1}^U(.) + (1-\delta_s) \Big(\\
& \pi_F(1-\pi_I) \max\Big\{ V_{t+1}^s(.,\theta), V_{t+1}^F(.,\theta') \Big\} \\
& + (1-\pi_F)\pi_I \max\Big\{ V_{t+1}^s(.,\theta), V_{t+1}^I(.,\theta') \Big\} \\
& + \pi_F \pi_I \max\Big\{ V_{t+1}^s(.,\theta), V_{t+1}^I(.,\theta'), V_{t+1}^F(.,\theta') \Big\} \\
& + (1-\pi_F)(1-\pi_I) V_{t+1}^s(.,\theta) \Big) \Big] \Bigg\}
\end{aligned}$$

$$\begin{aligned}
\text{s.t.} \qquad & c + a' = (1+r_t)a + y_t(s,e,\theta) + d_{i,t} \\
& a' \in [\underline{a}, \infty), \qquad \tilde{c} = c - v(h),
\end{aligned}$$

and for each unemployed worker is

$$\begin{aligned}
V_t^U(a,\beta) = \max_{c,a'} \Bigg\{ & \frac{c^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \Big[(1-\pi_F)(1-\pi_I) V_{t+1}^U(.) \\
& + \pi_F(1-\pi_I) \max\Big\{ V_{t+1}^U(.), V_{t+1}^F(.) \Big\} \\
& + (1-\pi_F)\pi_I \max\Big\{ V_{t+1}^U(.), V_{t+1}^I(.) \Big\} \\
& + \pi_F \pi_I \max\Big\{ V_{t+1}^U(.), V_{t+1}^I(.), V_{t+1}^F(.) \Big\} \Big] \Bigg\}
\end{aligned}$$

$$\begin{aligned}
\text{s.t.} \qquad & c + a' = (1+r_t)a + y_t(U) + d_{i,t} \\
& a' \in [\underline{a}, \infty).
\end{aligned}$$

Formal workers maximize their utility taking hours as given, so their labor disutility is internalized at the union level, not by individual households. However, informal hours are chosen and therefore included in the Euler Equation Given their decision, their consumption-equivalent income is

$$w^I e h_t - v(h_t) = w^I e h_t - \psi \frac{h_{i,t} h_{i,t}^{1/\varphi}}{1+\frac{1}{\varphi}} = \frac{w^I e h_t}{1+\varphi}.$$

Firms

In the firm side, there is a representative final-good producer that aggregates a continuum of $j \in [0,1]$ intermediate inputs produced by formal firms,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\varepsilon > 0$ is a constant substitution elasticity (CES).

Each intermediate good j is produced by a monopolistic competitive firm $y_{j,t} = Z_t l_{j,t}$. They hire labor at real wage w_t and set a constant markup of $\mu = \frac{\varepsilon}{\varepsilon-1}$. Hence, $P_t = \mu Z_t W_t$, and the real wage equals $w_t = \frac{Z_t}{\mu}$ at every date in steady-state. Firms' dividends are taxed at the same rate τ_ℓ as labor income, so $d_t = (1-\tau_\ell)(Y_t - w_t L_t)$, where $L_t = \mathbb{E}[l_{j,t}]$ is the aggregate formal labor.

We assume sticky prices following Rotemberg (1982), standard in Neo-Keynesian's literature. Such as Hagedorn, Manovskii, and Mitman (2019), the Phillips curve for price inflation is given by

$$\ln(1+\pi_t) = \kappa \left[\frac{w_t}{Z_t} - \frac{1}{\mu}\right] + \frac{\ln(1+\pi_{t+1})}{1+r_{t+1}} \cdot \frac{Y_{t+1}}{Y_t},$$

Additionally, there is a representative informal firm operating in a perfectly competitive market and producing under constant returns to scale, so $Y_t^I = w_t^I L_t^I$. The steady-state wage gap $\xi = w_t^I / w_t$ is calibrated to match PNAD Contínua data.

Unions

Nominal wages are assumed to be sticky as in Erceg, Henderson, and Levin (2000), where unions set nominal wages to maximize agent utility subject to Rotemberg (1982)'s adjustment costs. We assume that unions allocate all formal labor hours uniformly across agents, so $h_{i,t} = h_t$ (Auclert, Rognlie, and Straub 2025).

Also, as in Hagedorn, Manovskii, and Mitman (2019), union's objective is to maximize the utility of an agent with average consumption $\bar{C}_t = \int_0^1 c_{it}\, di$, discounted by the average discount factor $\bar{\beta} = \lambda_I \beta_L + (1-\lambda_I)\beta_H$ (Auclert, Rognlie, Souchier, et al. 2021). This results in a Phillips curve for wage inflation $\pi_t^w = (1+\pi_t) w_t / w_{t-1}$ given by

$$\ln(1+\pi_t^w) = \kappa_w \left[\psi \cdot h_t^{1/\varphi} \cdot \tilde{C}_t^{1/\gamma} - \frac{(1-\tau_\ell) w_t L_t}{\mu_w}\right] + \bar{\beta} \ln(1+\pi_{t+1}^w).$$

The Phillips Curve calculations for price and wage are found in the Appendix.

Government

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r_{t-1}^* + \phi \pi_{t-1}$, where r^* is the neutral real interest rate, and $\phi > 1$ (Taylor 1993).

In the governmental sphere, we have a government that purchase public goods, collects taxes, makes transfers, and incurs debt; therefore, the government's budget constraint is characterized by

$$G_t + T_t \cdot \mathrm{BF}_t + (1+r_t) B_{t-1} = B_t + \tau_t Y_t,$$

which means that, in steady-state, $G = \tau wL - rB - T \cdot \mathrm{BF}$.

For dynamic simulations, we intend to apply two regimes: one financed by debt, with T fixed and $B_t = (1+r)B + T \cdot \mathrm{BF} - \tau wL - G$, and another financed by taxes with B fixed and T variable. For now, we consider fixed T_t and B_t, treated as exogenous, with G_t adjusting to it.

Market Clearing

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). We define $\Lambda_t(s, \beta, e, a)$ as the household distribution over a given period of time. First, the asset market reaches equilibrium when household savings equal government bonds

$$A_t = \int a_{i,t} \, d\Lambda_t(s, \beta, e, a) = B_t;$$

the labor market clears when the share of employment in the workforce equals the firm's demand for labor

$$\begin{aligned} L_t &= \int \mathbb{1}\{s_{i,t} = F\} \cdot \theta^F_{i,t} \cdot e_{i,t} \cdot h_{i,t} \, d\Lambda_t(s, \beta, e, a); \\ L^I_t &= \int \mathbb{1}\{s_{i,t} = I\} \cdot \theta^I_{i,t} \cdot \cdot e_{i,t} \cdot h_{i,t} \, d\Lambda_t(s, \beta, e, a); \\ \text{and} \qquad & F_t + I_t + U_t = 1; \end{aligned}$$

the BF benefit clears when it is given to the eligible households

$$\mathrm{BF}_t = U_t + \int \mathbb{1}\{s_{i,t} = I, e_{i,t} < e^*\} \, d\Lambda(s, \beta, e, a);$$

and the goods market clear when $Y + Y^I = C + G$ in the steady-state. However, due to the GHH utility, we have $C = \tilde{C} + \int v(h) \, d\Lambda_t(.)$ and

$$\int \psi \frac{h_{i,t}^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}} \, d\Lambda_t(s, \beta, e, a) = \int \frac{\varphi h_{i,t} w^I_t e_{i,t}}{1+\varphi} \, d\Lambda_t(s, \beta, e, a) = \frac{\varphi w^I_t L^I_t}{1+\varphi} = \frac{\varphi}{1+\varphi} Y^I_t.$$

Thus,

$$Y_t + Y^I_t = \int \tilde{c}_{i,t} \, d\Lambda_t(s, \beta, e, a) + G_t = \tilde{C}_t + \frac{\varphi}{1+\varphi} Y_I + G_t.$$

2. Calibration

Data

To calibrate this model, we use Brazilian data and standard parameters in HANK Literature. Most of the data refers to their average values in 2025.

The labor market moments were obtained in Pesquisa Nacional por Amostra de

Domicílio Contínua (PNAD-C), an annual survey reporting formality status, wages, weekly worked hours, etc. This panel is interesting to match mainly the sector sizes and the worked hours, for both formal and informal households. However, since it is a survey, it might have under-reporting of informal firms.

Some general macroeconomic measures were also used, such as the Government Debt (in the Brazilian National Treasure), the interest rates, and the inflation, available in the Índice Brasileiro de Geografia e Estatística (IBGE) The Gini coefficient of wealth was obtained from the 2025 Global Wealth Report, from UBS. Bolsa Família data is available in Cadastro Único (CadÚnico), a platform with several statistics about the benefit.

In the future, we aim to gather information also in the Relação Anual de Informações Sociais (RAIS), a mandatory annual reporting of firms, but without informality information and only up to 2003, and Economia Informal Urbana (ECINF), a survey conducted by IBGE that focus on small business, specially informal ones.

Method and Calibration

The main method is the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, et al. 2021). The central idea is to summarize the model by a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous ones.

The method is built to find General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G}\, d\mathbf{Z} = -\mathbf{F}_{\mathbf{X}}^{-1}\mathbf{F}_{\mathbf{Z}}\, d\mathbf{Z}.$$

We are interested in calculating $dC = \mathbf{G}_{C,T}\, dT$, the total effect of the change in the conditional transfer on household consumption.

We resolve the heterogeneity by the Endogenous Grid Method (EGM) (Carroll 2006), which solves the value function iteration using the Euler equation, without solving the household maximization problem. The EGM was combined with the discrete choice among sectors using Iskhakov et al. (2017)'s methodology. Each model period stands for one quarter and the computational time horizon is 100 quarters (or 25 years).

The idiosyncratic productivity follows a log-normal $AR(1)$ process, $\ln e' = \rho_e \ln e + \varepsilon^e$, with $\varepsilon^e \sim N(0, \sigma_e^2)$. This is approximated by a finite Markov chain with N_e grid points (Rouwenhorst 1995), which is more accurate than Taucher for highly persistent processes.

We assume the sector productivity follows a log-normal $\ln \theta^s \sim N(\mu_s, \sigma_s^2)$, that is discretized by a Gauss-Hermite quadrature with N_θ nodes (Meghir, Narita, and Robin 2015). We also assume a log-spaced grid for assets, in order to generate more robustness near the credit constraint.

The discount factor transition (β_L, β_H) is modeled by a redrawing probability of $q = 0.10$, corresponding to an average type duration of 25 years, used to capture a generation (Auclert, Rognlie, and Straub 2025) This aims to capture similar results in

Table 1: External Calibration

	Description	Value	Source
Households			
γ	EIS	0.5	Standard in Literature
δ_β	Difference: $\beta_H - \beta_L$	0.12	Approximate Gini Coefficient
ω_I	Share of Impatient Agents	0.50	Auclert, Rognlie, and Straub (2025)
Productivity and Asset Grid			
$\underline{a}$	Borrowing constraint	0.0	No Borrowing (McKay and Reis 2016)
$\bar{a}$	Maximum asset	200	Sufficient to cover the top of dist.
N_a	Number of Asset Grid	200	Accuracy near constraint
ρ_e	Persistence of Log-Productivity	0.966	Kaplan, Moll, and Violante (2018)
σ_e	Productivity Std. Dev.	0.85	Bartal and Becard (2024)
N_e	Number of Nodes	15	Standard Discretization
μ_F, μ_I	Sector Average Productivity	0.0,−0.2	Standard Differentiation
σ_F, σ_I	Sector Std. Dev. Productivity	0.2,0.4	Standard Differentiation
N_θ	Number of Nodes	3	Standard Discretization
Firms			
Y	Aggregate Output	1.0	Normalization
ξ	Wage gap $= w^I/w$	0.64	PNAD-C Data
μ	Price Markup	1.11	Basu and Fernald (1997)
μ_w	Price Markup	1.11	Equal to μ
κ	Price-NKPC Slope	0.025	Hazell et al. (2022)
κ_w	Wage-NKPC slope	0.025	Equal to κ
Government and Monetary Policy			
r^*	Neutral Real Interest Rate	1.0%	4% annual average
ϕ	Taylor Rule Coefficient	1.5	Standard (Taylor 1993)
τ_ℓ	Labor Income Tax Rate	0.27	Average Tax Payroll in Brazil
T/w	Bolsa Família Transfer	0.14	CadÚnico, PNAD-C

Table 2: Endogenous Calibration

	Description	Value	Target	Model	Data	Source
Households						
ψ	Disutility of Labor	0.42	h_F	1.0	1.0	Normalization
φ	Frisch Elasticity	0.15	$\mathbb{E}(h_I)$	0.89	0.87	PNAD-C Data
β_H	Patient's Discount Factor	0.971	B/Y	3.2	3.2	Tesouro Nacional
Labor market						
π_F	Formal Offer Probability	0.10	F	44.5%	50.0%	PNAD-C Data
π_I	Informal Offer Probability	0.30	I	40.2%	44.2%	PNAD-C Data
δ_F, δ_I	Layoff Probabilities	0.03, 0.10	U	15.4%	5.8%	PNAD-C Data
$\bar{y}$	Bolsa Família Threshold	0.6	BF	0.433	0.294	CadÚnico, PNAD-C

relation to a life-cycle overlapping generations with permanent ability shock model.

We solve the steady-state equilibrium computationally by Brent's method, internally calibrating parameters in order to clear the asset market, the labor market, the government budget constraint, and the wage Phillips curve. The external calibration is shown in table 1 and the endogenous calibration is in 2. The goods market was left untargeted to verify the Walras' Law.

The output is normalized $Y = 1.0$ and we defined $\mu = 1.11$, which is standard in the HANK literature (Kaplan, Moll, and Violante 2018; Auclert, Rognlie, and Straub 2025) [3], and $\kappa = \kappa_w = 0.025$, consistent with Hazell et al. (2022) estimates.

Taylor Rule is pretty standard, with $\phi = 1.5$ and $r^* = 1\%$, consistent with a 4% annual real interest rate for Brazilian economy (Santos and Muinhos 2024). The Government debt is set to 80% of yearly GDP, consistent with Brazilian National Treasure data. Wage tax is an average of all contributions levied on the payroll, $\tau_\ell = 0.27$.

The Bolsa Família statistics were obtained from the Cadastro Único [4], where we calibrated the T/w ratio in steady-state to be equal to the data from the Brazilian economy, where $T =$ R$ 600 and $w =$ R$ 4,294. We also calibrated the Bolsa Família threshold $\bar{y}$ to match the actual size of the program [5].

[3] Markup $\mu = 1.11$ is also consistent with the estimates of Basu and Fernald (1997).

[4] Available in VIS DATA 3 and Cadastro Único.

[5] We used the average household size in PNAD-C to calculate a percentage estimate of the program.

3. Results

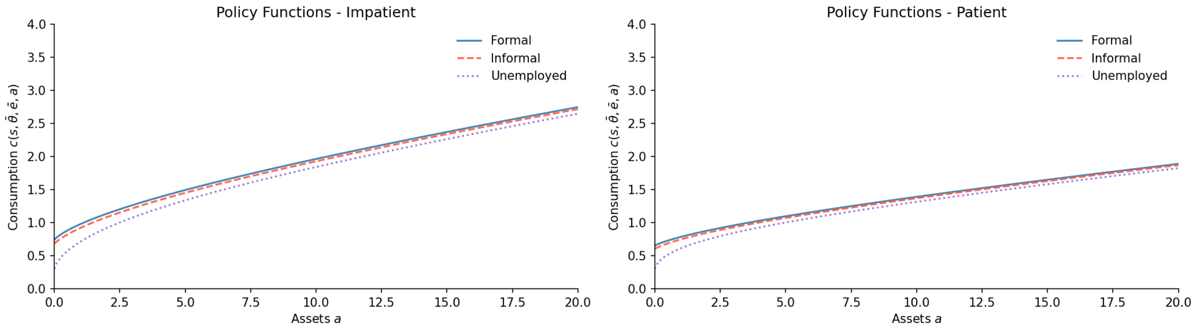

Figure 1: Steady-State Consumption Policy Function

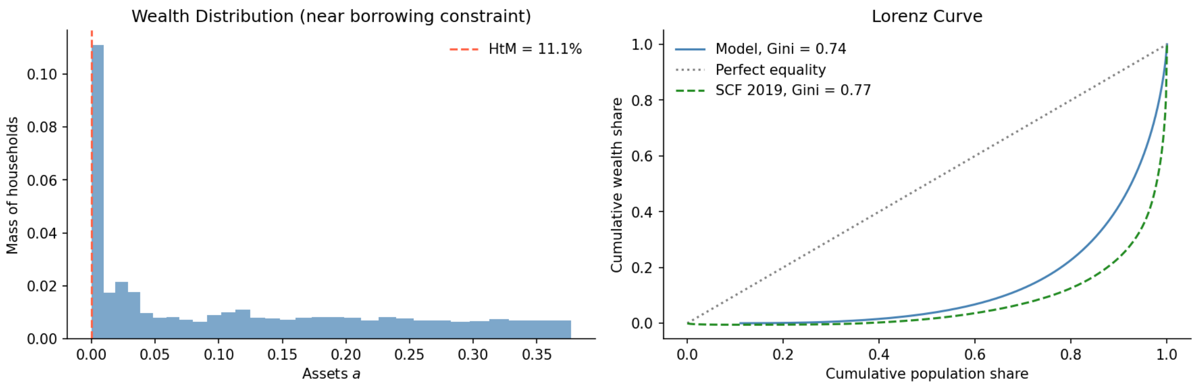

Figure 2: Wealth Distribution

Table 3: Steady state: Bolsa Família vs. no-transfer counterfactual

	With BF	No BF	Δ
Formal share α_F	44.5%	74.7%	-30.3
Informal share α_I	40.2%	15.0%	+25.2
Unemployment u	15.4%	10.3%	+5.1
Informality (of employed)	47.4%	16.7%	+30.8
Hand-to-mouth	11.1%	0.4%	+10.7
Wealth Gini	0.742	0.611	+0.131
Consumption Gini	0.399	0.446	-0.047
Aggregate consumption C	1.227	0.880	+0.348
Formal output Y	1.000	1.000	+0.000
BF spending	0.109	0.000	+0.109
Welfare $\mathbb{E}[V]$	-18.380	-29.697	+11.317

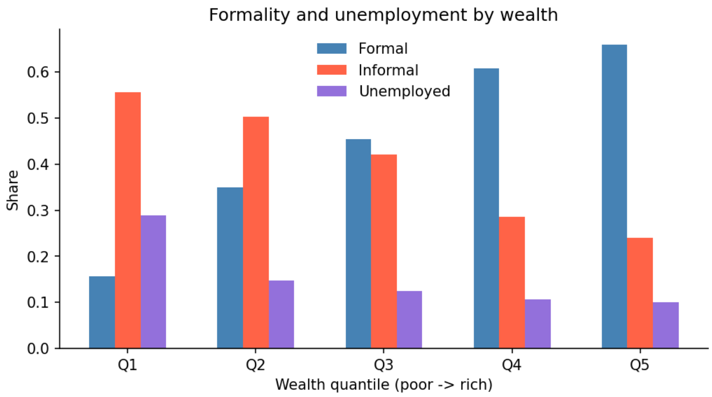

Figure 3: Formality and Unemployment by Wealth

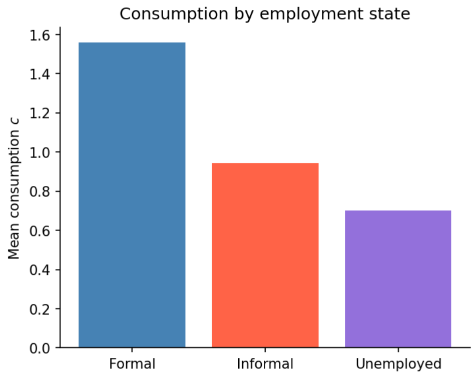

Figure 4: Steady-State Average Consumption by State

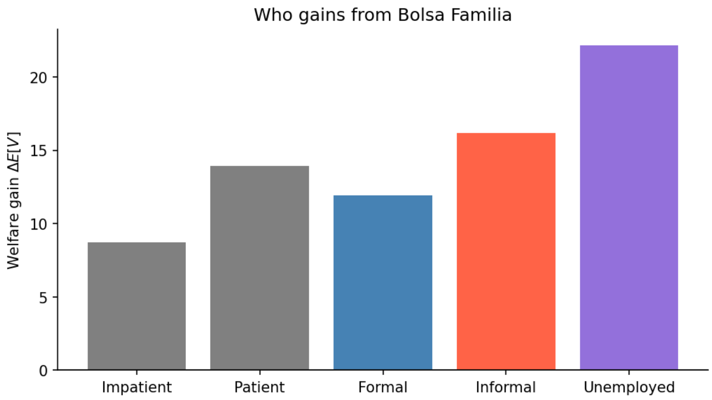

Figure 5: Welfare gains by Type of Household

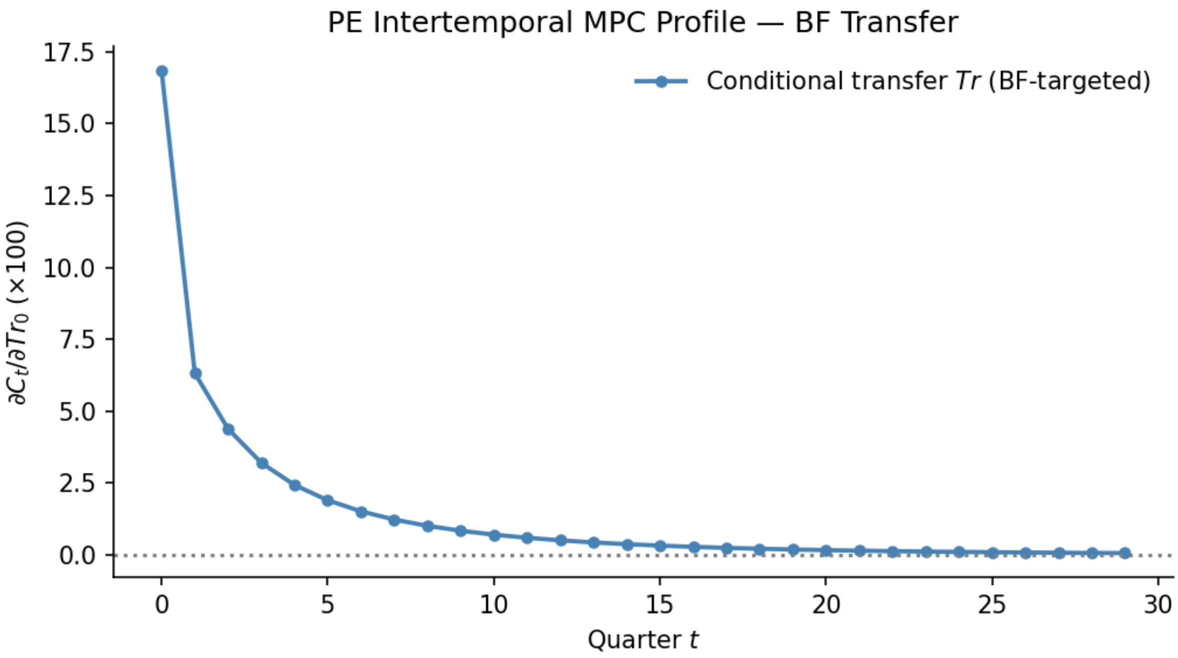

Figure 6: Partial Equilibrium MPC Profiles

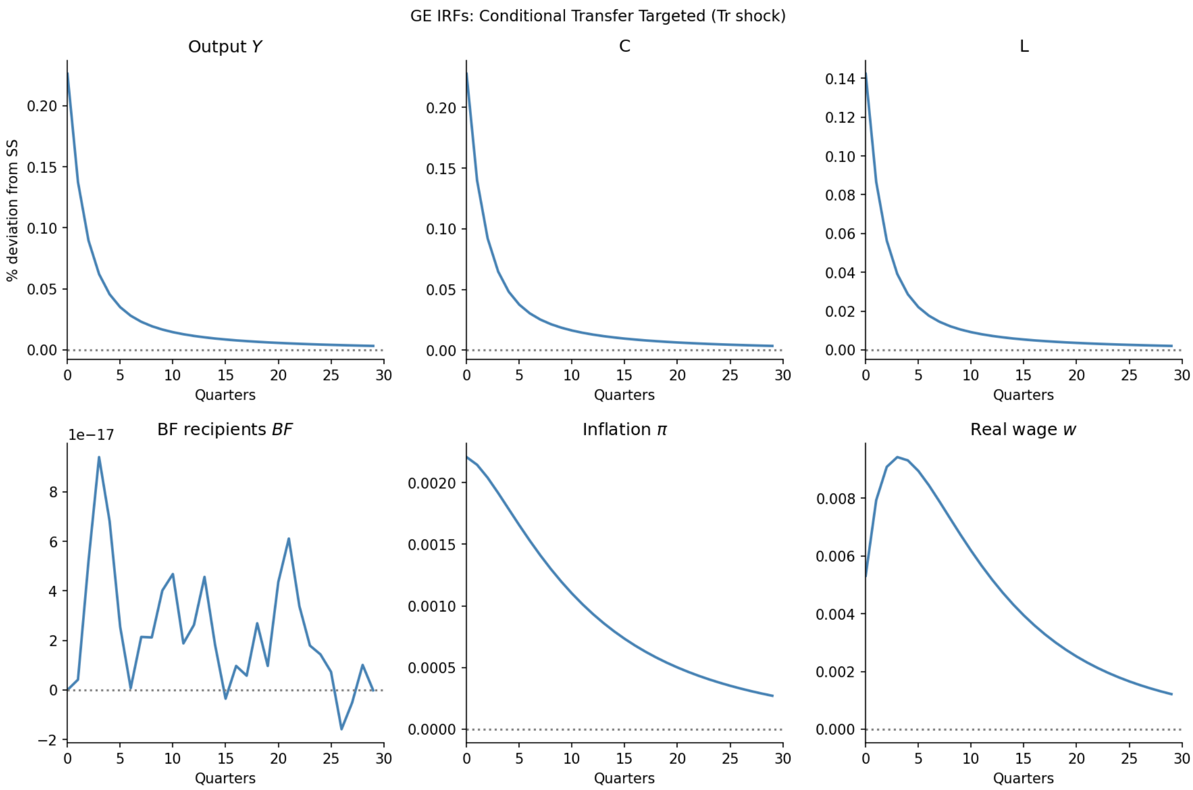

Figure 7: IRFs: BF-Targeted (shock in T)

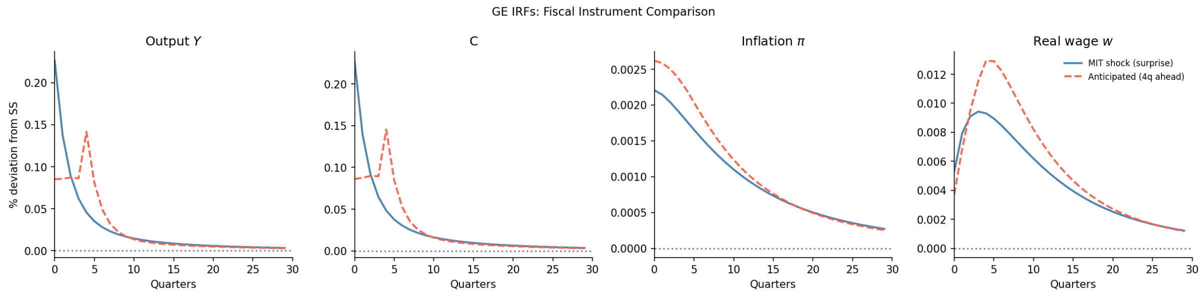

Figure 8: IRFs: Comparison (shock in T, MIT vs Antecipated)

Robustness Tests

Lastly, in order to verify whether this results are representatives and robust, we intent to do some robustness tests changing some calibration parameters, such as γ, δ_β, ω_I, ξ, μ, and κ_w. This parameters might have minor implications in the results, but it is expected that the model continues to have the same results.

Further, since there are several combinations of the sector transition probabilities that implies in the same sector sizes, we aim to changing each of this probability and capture the result. This change will affect the dynamics and may alter the transfer effects, due to a faster or slower transition to the informal sector. If the transition data is feasible in PNAD-C, it will not be necessary.

Also, as we commented early, we intent to perform other dividend specifications instead of $d_{i,t} = d_t \cdot e_{i,t}$. We also aim to change the GHH utility to a standard separable utility in the form of $u(c) - v(h)$, which must have similar results.

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Appendix

Price Phillips Curve

Let Y_t be the aggregate output of the economy, a CES of the intermediary inputs,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

This firm maximizes its output subject to $\int_0^1 p_{i,j} y_{i,j} \, dj$. The FOC implies that the demand for the intermediate good is given by

$$y_{j,t} = \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t,$$

where P_t is the equilibrium price of the final good,

$$P_t = \left(\int_0^1 p_{j,t}^{1-\varepsilon} \, dj \right)^{\frac{1}{1-\varepsilon}}.$$

We assume a Rotemberg (1982) quadratic price adjustment cost given by

$$\frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t,$$

where $\theta_p = (\varepsilon - 1)/\kappa$ is the adjustment cost multiplier and $\bar{\Pi}$ the steady-state inflation.

Let $\Xi(y)$ be the intermediary firm's cost. Given last period's price $p_{j,t-1}$, the firm chooses this period's price $p_{j,t}$ to maximize the present discounted values of future profits,

$$V_t^{IF}(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} y_{j,t} - \Xi(y_{j,t}) - \frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}^{IF}(p_{j,t}).$$

We can replace the demand function into this problem, so

$$V_t(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t - \Xi(y_{j,t}) - \frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}(p_{j,t}).$$

The FOC with respect to $p_{j,t}$ is

$$(1-\varepsilon) \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} \frac{Y_t}{P_t} + \varepsilon mc_{j,t} - \theta_p \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right) \frac{Y_t}{p_{j,t}} + \frac{1}{1+r_{t+1}} V_{t+1}'(p_{j,t}) = 0,$$

where $mc_t = \partial \Xi / \partial y_{j,t}$, and the envelope condition gives us

$$V_{t+1}' = \theta_p \left(\frac{p_{j,t+1}}{p_{j,t}} - \bar{\Pi} \right) \frac{p_{j,t+1}}{p_{j,t}} \frac{Y_{t+1}}{p_{j,t}}.$$

Thus, taking $p_{j,t} = P_t$ in equilibrium,

$$\theta_p(\pi_t - \bar{\Pi})\pi_t = \varepsilon \left(mc_t - \frac{\varepsilon - 1}{\varepsilon} \right) + \frac{1}{1+r_{t+1}} \theta_p(\pi_{t+1} - \bar{\Pi})\pi_{t+1} \frac{Y_{t+1}}{Y_t},$$

where we define the firm's markup as $\mu = \frac{\varepsilon}{\varepsilon-1}$. Linearizing around the steady-state $\pi_t = \bar{\Pi}$,

$$\ln(1+\pi_t) = \frac{\varepsilon}{\theta_p}\left(mc_t - \frac{1}{\mu}\right) + \frac{1}{1+r_{t+1}}\ln(1+\pi_{t+1})\frac{Y_{t+1}}{Y_t},$$

where $\kappa = \varepsilon/\theta_p$ and $mc_t = w_t/Z_t$. This environment is equivalent to Calvo (1983) in first order around zero inflation.

Wage Phillips Curve

We will use the same setting as Hagedorn, Manovskii, and Mitman (2019), but considering just the formal sector. Assume there is a union that aggregate effective labor input into

$$h_t = \left(\int_0^1 e_{i,t}(h_{i,t})^{\frac{\varepsilon_w-1}{\varepsilon_w}}\, di\right)^{\frac{\varepsilon_w}{\varepsilon_w-1}},$$

sells households labor services $e_{i,t}h_{i,t}$ at nominal wage $W_{i,t}$ to the labor recruiter, which minimizes costs $\int_0^1 W_{i,t}e_{i,t}h_{i,t}\, di$, implying a demand of

$$h_{i,t} = \left(\frac{W_{i,t}}{W_t}\right)^{-\varepsilon_w} h_t.$$

The unions sets a nominal wage $\hat{W}_t$ for an effect unit of labor, $W_{i,t} = \hat{W}_t$ to maximize profits subject to Rotemberg (1982)'s wage adjustment costs

$$e_{i,t}\frac{\theta_w}{2}\left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w\right)^2 h_t.$$

We assume that the union's wage setting problem is

$$V_t^w = \max_{\hat{W}_t}\int\left[\frac{(1-\tau_\ell)\hat{W}_t}{P_t}e_{i,t}h_{i,t} - \frac{v(h_{i,t})}{u'(\tilde{C}_t)} - e_{i,t}h_t\frac{\theta_w}{2}\left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w\right)^2\right]d\Lambda + \bar{\beta}V_{t+1}^w.$$

Using the same algebra as the price Phillips curve, assuming that the union allocates the same amount of hours in each worker $h_{i,t} = h_t$, and using $\int_F e_{i,t}d\Lambda = L_t$, it is possible to find that

$$\theta_w(\pi_t^w - \bar{\Pi}^w)\pi_t^w = \varepsilon_w\left[\frac{v'(h_t)}{u'(\tilde{C}_t)} - \frac{\varepsilon_w-1}{\varepsilon_w}(1-\tau_\ell)w_tL_t\right] + \bar{\beta}\theta_w(\pi_{t+1}^w - \bar{\Pi}^w)\pi_{t+1}^w\frac{h_{t+1}}{h_t},$$

which we can linearize around the steady state $\pi_t^w = \bar{\Pi}^w$ and take $h_{t+1} = h_t$, since there is no population nor technological growth. Thus,

$$\ln(1+\pi_t^w) = \frac{\varepsilon_w}{\theta_w}\left[\frac{v'(h_t)}{u'(\tilde{C}_t)} - (1-\tau_\ell)w_tL_t\frac{\varepsilon_w-1}{\varepsilon_w}\right] + \bar{\beta}\ln(1+\pi_{t+1}^w).$$

Therefore, defining $\kappa_w = \varepsilon_w/\theta_w$ and $\mu_w = \varepsilon_w/(\varepsilon_w-1)$, the wage Phillips curve is

$$\ln(1+\pi_t^w) = \kappa_w\left[\psi\cdot h_t^{1/\varphi}\cdot\tilde{C}_t^{1/\gamma} - \frac{(1-\tau_\ell)w_tL_t}{\mu_w}\right] + \bar{\beta}\ln(1+\pi_{t+1}^w).$$